



University of Colorado
Denver | Anschutz Medical Campus

Department of Computer Science and Engineering
College of Engineering and Applied Science
University of Colorado Denver

COURSE SYLLABUS

Course Number: CSCI 4230-5230

Course Title: *Natural Language Processing and Generative AI (NLP & GenAI)*

Instructor:	<i>Farnoush Banaei-Kashani</i>	Term:	<i>Fall 2026</i>
Office:	<i>LW 803</i>	Class Meeting Days:	<i>N/A</i>
Phone:	<i>303-315-0116</i>	Class Meeting Hours:	<i>N/A</i>
Email:	<i>farnoush.banaei-kashani@ucdenver.edu</i>	Class Location:	<i>Online</i>
Website:	http://cse.ucdenver.edu/~farnoush	Lab Location:	<i>N/A</i>
Office Hours:	<i>Mondays 5:00pm-8:30pm and Thursdays 6:00pm-7:30pm (<u>by appointment only</u> - to make appointment stop by the department front desk in person, call the department at 303-315-1408, or email the department at ComputerScience@ucdenver.edu)</i>		

COURSE OVERVIEW

I. Welcome!

Natural Language Processing (NLP) is a crucial component of artificial intelligence (AI), modeling how people share information. In recent years, deep learning approaches (including Generative AI methods, exemplified by ChatGPT) have obtained groundbreaking performance on many NLP tasks. In this course, students gain a thorough introduction to cutting-edge machine learning and deep learning computing models for NLP.

II. University Course Catalog Description

Introduces advanced machine/deep learning as well as generative AI models and techniques for natural language processing (NLP).

III. Course Overview

Natural language processing (NLP) is one of the most important technologies of the information age. Applications of NLP are everywhere because people communicate almost everything in language: web search, advertising, emails, customer service, language translation, virtual agents, medical reports, politics, etc. In the last decade, deep learning approaches have obtained very high performance across many different NLP tasks, using single end-to-end neural models that do not require traditional, task-specific feature engineering. In this course, students will gain a thorough introduction to cutting-edge machine learning, deep learning and Generative AI models for NLP. Through lectures, hands-on assignments and a term project, students will learn the necessary skills to design, implement, and understand their own computational models for NLP tasks.

IV. Course Goals and Learning Objectives

In this course, NLP concepts are mainly taught through lectures on the selected topics. In particular, we follow topics under the following areas:

- Word Vectors, Word Window Classification, Language Models
- Dependency Parsing
- Recurrent Neural Networks and Language Models
- Seq2Seq and Subword Models
- Machine Translation, Self-Attention and Transformers
- Pre-training
- Prompting and Reinforcement Learning from Human Feedback
- Natural Language Generation

In addition to lectures, students will practice how to utilize the learned concepts through five programming assignments that focus on the use of NLP methods with real datasets from various application domains to address real-world problems.

Finally, to stimulate creativity of the students in the context of natural language processing and to enhance their research skills, an individual or group of (up to 4) students will take on a term project, with which throughout the semester the student(s) will work toward introduction and implementation of novel solutions for their proposed NLP problems. **The project is mandatory for students enrolled in CSCI 5230 section and optional (for extra credit) for students enrolled in the CSCI 4230 section of the course.** The students are allowed to introduce their own project ideas, or alternatively communicate with the instructor of the course to identify relevant project ideas in the context of existing research projects. As a result of the term project, the student(s) are expected to develop and demonstrate an interesting proof-of-concept data scientific system prototype, and will prepare a written report as well as power-point and video presentation explaining their innovation. See the course Canvas site to learn more about project options and requirements.

V. Course Prerequisites

As a graduate-level course, this course does not enforce any prerequisites. However, the course is recommended for student that have sufficient familiarity with the following concepts:

- College Calculus, Linear Algebra
- Basic Probability and Statistics
- Basic Machine Learning and Deep Learning Knowledge: We will be formulating cost functions, taking derivatives and performing optimization with gradient descent. If you already have basic machine learning and/or deep learning knowledge, the course will be easier; however it is possible to take this course without it. There are many introductions to ML, in webpage, book, and video form. One approachable introduction is Hal Daumé's "[A Course in Machine Learning](#)". Reading the first 5 chapters of this book would be good background. Knowing the first 7 chapters would be even better!
- Programming Knowledge: It is assumed that students are skilled programmers, and also can learn to program in a new programming or script language on their own in a short timeframe, need be. We will mainly use Python and work with Pytorch and NumPy for programming purposes. We will provide self-paced tutorials for self-study for those who may need to brush up on their Python programming skills, or are new to Python but have sufficient programming proficiency to pick it up quickly with self-study.

VI. Course Credits

3 credit hours

VII. Required Textbook

N/A

VIII. Supplementary (Optional) Materials

The following texts are useful, but none are required. All of them can be read free online.

- Dan Jurafsky and James H. Martin. [Speech and Language Processing \(3rd ed. draft\)](#)
- Jacob Eisenstein. [Natural Language Processing](#)
- Ian Goodfellow, Yoshua Bengio, and Aaron Courville. [Deep Learning](#)
- Lewis Tunstall, Leandro von Werra, and Thomas Wolf. [Natural Language Processing with Transformers](#)

If you have no background in neural networks but would like to take the course anyway, you might well find one of these books helpful to give you more background:

- Michael A. Nielsen. [Neural Networks and Deep Learning](#)
- Eugene Charniak. [Introduction to Deep Learning](#)

Course Schedule

Week	Topic	Assignments	Project
1	Introduction		
2	Lecture 1 – Word Vectors		
3	Lecture 2 – Word Vectors (cont'd), Word Senses, and Neural Classifiers	#1 Assigned on 9/4	
4	Lecture 3 – Neural Networks		
5	Lecture 4 – Dependency Parsing	#1 Due & #2 Assigned on 9/18	
6	Lecture 5 – Language Models, Recurrent Neural Networks		Proposal Due on 9/25
7	Lecture 6 – Recurrent Neural Networks (cont'd), Neural Machine Translation	#2 Due & #3 Assigned on 10/2	
8	Lecture 7 – Neural Machine Translation (cont'd), Sequence-to-Sequence Models, Attention		
9	Lecture 8 – Self-Attention and Transformers – Part I		
10	Lecture 8 – Self-Attention and Transformers – Part II	#3 Due & #4 Assigned on 10/23	
11	Lecture 9 – Pretraining – Part I		
12	Lecture 9 – Pretraining – Part II		

13	Lecture 10 – Prompting and RLHF – Part I	#4 Due & #5 Assigned on 11/13	
14	Lecture 10 – Prompting and RLHF – Part II		
15	FALL BREAK		
16	Lecture 11 – Neural Language Generation	#5 Due on 12/4	Final Report Due on 12/4 & Presentation Due on 12/4
Finals Week	Final Exam (in-person; time and location TBA)		

EVALUATION

X. Assignments

In this course, we administrate one exam (final exam) to assess students' understanding of the presented concepts. In addition, hands-on skills of the students in use and design of NLP solutions are evaluated with 5 programming assignments as well as a term project (NB: the term project is optional for CSCI 4230 students).

XI. Basis for Final Grade

In this course the grading policy for CSCI 5230 students is as follows:

Assessment	Points Possible	Percent of Final Grade
Assignments #1 to #5	100 (each)	8% (each)
Term Project	100	25%
Final Exam	100	35%
		100%

In this course the grading policy for CSCI 4230 students is as follows:

Assessment	Points Possible	Percent of Final Grade
Assignments #1 to #5	100 (each)	12% (each)
Optional Term Project	100	10%
Final Exam	100	40%
		110%

Also the adopted grading scale (based on percentage grade) is as follows:

Grading Scale (points or %)	
94-100	A
90-93	A-
87-89	B+
84-86	B
80-83	B-
77-79	C+
74-76	C

70-73	C-
67-69	D+
64-66	D
60-63	D-
0 - 59	F

XII. Grade Dissemination

Grades for term project and exams will be announced via the Canvas shell of the course. Students will be given a chance discuss their grades with the TA/instructor for any required adjustment. Once all grades assessments are finalized, the final letter grade of the students will be calculated based on the approach detailed under Section XI above, and delivered to the registrar office to be posted.

FYI, CU Denver utilizes web grading which is accessed through UCDAccess. All web grading information can be found at www.ucdenver.edu/student-services/resources/registrar/faculty-staff/

COURSE PROCEDURES

XIII. Course Policies: Grades

Attendance Policy: N/A (online course)

Late Work Policy: There are no make-ups for deliverables and late submissions will NOT be considered.

Extra Credit Policy: Instructor will decide whether to include any extra credit, should there be any need. Any extra credit points you may be granted are added as additional points to your total grade. Any points over 100% are not counted.

Grades of "Incomplete": The current university policy concerning incomplete grades will be followed in this course. Incomplete grades are given only in situations where unexpected emergencies prevent a student from completing the course and the remaining work can be completed the next semester. Your instructor is the final authority on whether you qualify for an incomplete. Incomplete work must be finished by the end of the subsequent semester or the "I" will automatically be recorded as an "F" on your transcript.

Rewrite/Resubmit Policy: No rewrite/resubmission is allowed.

Group Work Policy: The term projects can be performed by a team of students. In group projects, the contribution of each individual student to the project is the sole measure used for evaluation of the student performance in the term project.

XIV. Course Policies: Technology and Media

Email: Students can communicate with the instructor and the TA via their official UCDenver email address only. You should allow 24 hours (or more during the peak submission periods and weekends) to receive a response. All technical questions about the course materials should first be addressed to the TA; if students are not satisfied with the received answer, they can then escalate their questions to be considered by the instructor.

Canvas: All course materials, updates, and communications to students are and will be posted on the Canvas site of the course. It is your responsibility to set your Canvas notification policy to "ASAP" for all notifications/announcements issued by instructor and TA, as such announcements (sometimes time-sensitive) may not be accessible in any other way.

Communications between Students: Students will use the course forum account to exchange thoughts on labs, assignments, term projects, and lectures. TA may monitor the discussions on assignments and may contribute to the on-going discussions, but for any specific questions for TA, students must directly contact the TA via her/his contact posted on the Canvas shell of the course.

Library-supplied online databases and collections: You should make yourself familiar with the procedure to access and use books and publication databases provided to you by the campus library. In particular, the proposed supplementary materials are available at the Auraria Library and many relevant research publications are also accessible in either ACM or IEEE libraries/databases through the same. Here is the contact info for the Auraria Library librarians, should you have any questions:
<http://library.auraria.edu/services/researchhelp/>

XV. Course Policies: Student Expectations

Civility: My commitment is to create a climate for learning characterized by respect for each other and the contributions each person makes to class. I ask that you make a similar commitment.

Campus Assessment, Response & Evaluation (CARE)

Campus Assessment, Response & Evaluation (CARE) Team, which was created at the University of Colorado Denver and Anschutz Medical Campuses to address the health and safety needs of students as well as the campus community. The purpose of the team is to assess whether individuals pose a risk to themselves or others and to intervene when necessary and, more generally, to identify and provide assistance to those in need. The team takes a preventive approach to risk assessment by offering resources, referrals, and support to both the concerning individual and those impacted by their behavior. <http://www.ucdenver.edu/CARE>

Religious Observances: Should there be a conflict between the normal class schedule and major religious observances, you are expected to inform the instructor in advance (2 weeks at least), if you intend to miss class to observe a holy day of your religious faith.

UNIVERSITY POLICIES

XVI. Access

Disability Access: The University of Colorado Denver is committed to providing reasonable accommodation and access to programs and services to persons with disabilities. Students with disabilities who want academic accommodations must register with Disability Resources and Services (DRS) in Academic Building 1, #2116, Phone: 303- 315-3510 , Fax: 303-315- 3515. I will be happy to provide approved accommodations, once you provide me with a copy of DRS's letter. [DRS requires students to provide current and adequate documentation of their disabilities. Once a student has registered with DRS, DRS will review the documentation and assess the student's request for academic accommodations in light of the documentation. DRS will then provide the student with a letter indicating which academic accommodations have been approved.]

XVII. Academic Honesty

Student Code of Conduct: Students are expected to know, understand, and comply with the ethical standards of the university, including rules against plagiarism, cheating, fabrication and falsification, multiple submissions, misuse of academic materials, and complicity in academic dishonesty. In particular, the Honor Code outlined below is the College of Engineering and Applied Science statement on academic integrity. The Code articulates the College's expectations of its students and faculty in establishing and maintaining the highest standards in academic work. The complete text of the honor code is available here and all students MUST read and adhere to this code:

https://engineering.ucdenver.edu/docs/librariesprovider29/default-document-library/student-honor-code-revised12_2022.pdf?sfvrsn=8624fbb_0

Below please find more detailed definitions of conducts that infringe the Student Honor Code:

Plagiarism is the use of another person's ideas or words without acknowledgement. The incorporation of another person's work into yours requires appropriate identification and acknowledgement. Examples of plagiarism when the source is not noted include: word-for-word copying of another person's ideas or words; the "mosaic" (interspersing your own words here and there while, in essence, copying another's work); the paraphrase (the rewriting of another's work, while still using their basic ideas or theories); fabrication (inventing or counterfeiting sources); submission of another's work as your own; and neglecting quotation marks when including direct quotes, even on material that is otherwise acknowledge.

Cheating involves the possession, communication, or use of information, materials, notes, study aids, or other devices and rubrics not specifically authorized by the course instructor in any academic exercise, or unauthorized communication with any other

person during an academic exercise. Examples of cheating include: copying from another's work or receiving unauthorized assistance from another; using a calculator, computer, or the internet when its use has been precluded; collaborating with another or others without the consent of the instructor; submitting another's work as one's own.

Fabrication involves inventing or counterfeiting information—creating results not properly obtained through study or laboratory experiment. Falsification involves deliberate alteration or changing of results to suit one's needs in an experiment or academic exercise.

Multiple submissions involves submitting academic work in a current course when academic credit for the work was previously earned in another course, when such submission is made without the current course instructor's authorization.

Misuse of academic materials includes: theft/destruction of library or reference materials or computer programs; theft/destruction of another student's notes or materials; unauthorized possession of another student's notes or materials; theft/destruction of examinations, papers, or assignments; unauthorized assistance in locating/using sources of information when forbidden or not authorized by the instructor; unauthorized possession, disposition, or use of examinations or answer keys; unauthorized alteration, forgery, fabrication, or falsification of academic records; unauthorized sale or purchase of examinations, papers, or assignments.

Complicity in academic dishonesty involves knowingly contributing to or cooperating with another's act(s) of academic dishonesty.

XVIII. Important Dates to Remember

Academic Calendar: University's current Academic Calendar for Fall 2026 is available here for your reference:

<https://catalog.ucdenver.edu/cu-denver/graduate/records-registration/registration/academic-calendars/>